3.2.1.2 Helical gear

It is also a circular shaped metal made on a cylindrical metal having the teeth cut not parallel to the axis but in helical shaped cut at certain angle inclined to the cylinder's axis. This type of gear is called as the helical gear.



Figure 3.2.1.2 Helical gear

We can transmit the rotational power to two shafts which are either parallel or non parallel through the helical gear. Compared to straight spur gears more power can be transmitted through helical gears. This type of gear is used in synchromesh gear box and constant mesh gear box.

3.2.1.3 Bevel gear

In this gear the teeth are cut on the top surface of the shaft in tapper. This type of gear is generally used to connect the shafts which are perpendicular to each other and transmit the power in the perpendicular i.e. at 90° angle direction. Bevel gear is capable of withstanding more power than other gears. The bevel gear is used in differential, pinion, crown, wheel, planet and sun gears.



Figure 3.2.1.3 Bevel gear

3.2.1.4 Rack and pinion gear

In this arrangement teeth are cut on a long flat metal bar called as rack. A small wheel called as pinion is in mesh with rack. In this way the rotating power of the pinion is transformed into a straight line or reciprocating motion. This type of gears are used in the gear boxes in the steering system and in many mechanical machines.



Figure 3.2.1.4 Rack and pinion gear

3.2.1.5 Worm and worm gear

It is used to transmit the rotational power by connecting two shafts that are

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